IIT MADRAS — PhD THESIS RESEARCH

SAMBP: Smart Adaptive Model-Based Protection

Research Gap-Filling Plan

Novel Research Agenda

TR-56 through TR-67

April 2026

Prepared for: PhD Thesis and Publication Pipeline

12 Proposed Technical Reports · 9 Journal Papers · 6 Patents · 5 Conference Papers

Contents

Executive Summary and Priority Ranking

Context and Starting Point

The SAMBP (Smart Adaptive Model-Based Protection) project developed a unified inverse-estimation framework for power system protection across 53 technical reports (TR-03 to TR-55). The core innovation is fitting a physics-based parametric model to measured fault current waveforms using Levenberg–Marquardt (LM) optimisation, then gating trip decisions on estimation confidence metrics (κ_n, f_{int}, γ).

The existing SAMBP framework addressed:

- **87L** line differential (SG and IBR networks) — most comprehensive coverage (TR-03 to TR-27)
- **87B** busbar differential — full topology coverage including GOOSE reconfiguration
- **87T** transformer differential — foundational framework only (TR-04)
- **21** distance relay — SG Mho and IBR cross-polar / quadrilateral / POTT
- **67N/67Q** directional overcurrent — IBR-specific
- **Adaptive OC 50/51** — synchronous generators and IBR microgrids

The coverage gap: All SAMBP protection was developed with either (a) a synchronous generator (SG) source or (b) a generic inverter-based resource (IBR) source. No work specifically addressed DFIG wind generators, solar PV inverters, transformer protection under IBR, generator protection relay elements (40, 78, 64G, 81), or the 87G generator differential for any generator type.

Prioritisation Principle

Research items are ranked by *mathematical novelty and disruption to existing protection standards*. An item scores highest if it:

1. Invalidates a current IEC/IEEE standard assumption in an IBR context
2. Introduces a genuinely new mathematical method to protection engineering
3. Has high commercial IP potential (replaces relay hardware or changes relay settings)

Master Priority Table

#	TR	Research Topic	Core Mathematical Challenge	Novelty
1	TR-56	DFIG fault model + inverse estimation	Piecewise-ODE with unknown crowbar breakpoint; smoothed Heaviside LM	■■■■■
2	TR-57	Bayesian SPRT for 87T inrush / fault	3-hypothesis sequential probability ratio test on waveform asymmetry	■■■■■
3	TR-58	Out-of-step (78) for hybrid SG–IBR	Generalised equal-area criterion; Lyapunov function for coupled ODEs	■■■■■
4	TR-59	Virtual 87G-DFIG using EKF observer	Slip-frequency EKF rotor current estimation without rotor CTs	■■■■□
5	TR-60	Loss-of-excitation (40) with IBR	Nonlinear IBR admittance correction to classical Mho circles	■■■■□
6	TR-61	Stator earth fault (64G) via injection	Optimal sub-harmonic injection frequency from SNR maximisation	■■■■□
7	TR-62	Solar PV sparse inverse model	2-parameter zero-DC model; k_{ibr}-driven model selection	■■■□□
8	TR-63	Under/over-frequency (81) zero-inertia	Inertia-normalised frequency deviation index for GFM networks	■■■□□
9	TR-64	87T IBR system integration	SPRT + sparse model integration across all transformer topologies	■■■□□
10	TR-65	Complete generator protection relay	Integration of TR-56–TR-63 into unified relay specification	■■□□□
11	TR-66	Monte Carlo robustness study	10,000-trial parametric validation of all new elements	■■□□□
12	TR-67	HIL validation protocol extension	Hardware-in-the-loop with DFIG and PV emulators	■■□□□

Phase 1: Foundational Mathematics (Priority 1–3)

TR-56: Piecewise-ODE Fault Model for DFIG with Unknown Crowbar Breakpoint

Research Gap — TR-56

All existing SAMBP inverse-estimation models (TR-03, sync_oc) assume a **continuously differentiable** fault current governed by a single ODE system. The DFIG Type-3 wind generator violates this assumption: when rotor current exceeds the crowbar threshold I_{cb}, the crowbar fires and **discontinuously inserts** resistance R_{cb} into the rotor circuit. The fault current transitions at an *a priori unknown* switching time t_{cb}. **No existing protection inverse-estimation framework handles unknown breakpoints in piecewise-ODE systems.**

Mathematical Formulation

Pre-crowbar fault current ($t_0 \leq t < t_{\mathrm{cb}}$)

$$i_a(t) = I_{\mathrm{ss}} \sin(\omega_s t + \phi_0) + I_{\mathrm{nat}} \sin((1-s)\omega_s t + \psi) \exp\left(-\frac{t-t_0}{\tau_r'}\right) + I_{\mathrm{dc}} \exp\left(-\frac{t-t_0}{\tau_s'}\right) \sin(\phi_0) \tag{2.1}$$

where $\tau_r' = L_r'/R_r \approx 0.5$–$2\,\mathrm{s}$ and $\tau_s' = L_s'/R_s \approx 20$–$200\,\mathrm{ms}$.

Post-crowbar model ($t \geq t_{\mathrm{cb}}$)

$$i_a(t) = I_{\mathrm{ss}} \sin(\omega_s t + \phi_{\mathrm{cb}}) + I''_{\mathrm{nat}} \sin((1-s)\omega_s t + \psi_{\mathrm{cb}}) \exp\left(-\frac{t-t_{\mathrm{cb}}}{\tau_r''}\right) + I''_{\mathrm{dc}} \exp\left(-\frac{t-t_{\mathrm{cb}}}{\tau_s''}\right) \sin(\phi_{\mathrm{cb}}) \tag{2.2}$$

where $\tau_r'' = L_r'/(R_r + R_{\mathrm{cb}}) \ll \tau_r'$ (crowbar collapses τ_r' from $\sim 1\,\mathrm{s}$ to 5–30 ms).

Full DFIG parameter vector

$$\boldsymbol{\theta}_{\mathrm{DFIG}} = \left[t_0,\ t_{\mathrm{cb}},\ I_{\mathrm{ss}},\ I_{\mathrm{nat}},\ \tau_r',\ \tau_s',\ I''_{\mathrm{nat}},\ \tau_r'',\ \tau_s'',\ \phi_0\right]^\top \in \mathbb{R}^{10} \tag{2.3}$$

Novel mathematical problem: unknown breakpoint t_{cb}

The cost function $F(\boldsymbol{\theta}_{\mathrm{DFIG}}) = \|\boldsymbol{r}(\boldsymbol{\theta}_{\mathrm{DFIG}})\|^2$ is **non-smooth** in t_{cb} because the model switches discretely. Standard LM cannot converge.

Proposed solution — Smoothed Heaviside continuation:

$$h_\varepsilon(t,\, t_{\mathrm{cb}}) = \sigma\!\left(\frac{t - t_{\mathrm{cb}}}{\varepsilon}\right) = \frac{1}{1 + \exp(-(t - t_{\mathrm{cb}})/\varepsilon)} \tag{2.4}$$

$$\hat{i}(t;\, \boldsymbol{\theta}, \varepsilon) = [1 - h_\varepsilon] \cdot i_{\mathrm{pre}}(t;\, \boldsymbol{\theta}_{\mathrm{pre}}) + h_\varepsilon \cdot i_{\mathrm{post}}(t;\, \boldsymbol{\theta}_{\mathrm{post}}) \tag{2.5}$$

$$\frac{\partial \hat{i}}{\partial t_{\mathrm{cb}}} = -\frac{1}{\varepsilon}\, h_\varepsilon (1 - h_\varepsilon)\, [i_{\mathrm{post}} - i_{\mathrm{pre}}] \quad \longrightarrow \quad \text{continuous} \;\Rightarrow\; \text{LM converges} \tag{2.6}$$

Continuation schedule: $\varepsilon \in \{1\,\mathrm{ms},\; 0.5\,\mathrm{ms},\; 0.2\,\mathrm{ms},\; 0.1\,\mathrm{ms}\}$ — anneal toward hard switch.

Crowbar initialisation via CUSUM changepoint detection

$$C_k = \max\!\left(0,\, C_{k-1} + \left|\frac{\mathrm{d}i_k}{\mathrm{d}t}\right| - \mu_0 - \kappa\right), \qquad \hat{t}_{\mathrm{cb}} = k \cdot T_s \;\; \text{when} \;\; C_k > h \tag{2.7}$$

DFIG-specific confidence gate

$$\kappa_n(\mathbf{J}_{\mathrm{DFIG}}) \lesssim 120 \quad (\text{vs. } 50 \text{ for 87L}) \quad \Rightarrow \quad \kappa_{\max} = 100 \tag{2.8}$$

$$f_{\mathrm{int,DFIG}} = 1 - w_1 \cdot \frac{|I''_{\mathrm{nat}} - I_{\mathrm{nat,exp}}(k_{\mathrm{ibr}})|}{I_{\mathrm{nat,exp}}} - w_2 \cdot \max\!\left(0,\, \frac{\tau''_r - \tau_{r,\mathrm{cb,max}}}{\tau_{r,\mathrm{cb,max}}}\right) \tag{2.9}$$

IP — Patent Family P1

Claim: A protection method for Type-3 DFIG wind generators comprising: (i) a piecewise-continuous 10-parameter fault current model with trainable crowbar switching time t_{cb}; (ii) a smoothed-Heaviside Levenberg–Marquardt estimator with annealing continuation schedule; (iii) a CUSUM-based crowbar detection initialiser; and (iv) a confidence gate conditioned on post-switching condition number κ_n and physics constraint $\tau''_r < \tau'_r/3$ — enabling selective DFIG protection without dependency on k_{ibr}.

Publications — TR-56

J1 (Primary): IEEE Transactions on Energy Conversion
"Inverse-Estimation-Based Protection for DFIG Wind Generators: Piecewise-ODE Model with Unknown Crowbar Breakpoint"

C1: IEEE PES General Meeting 2027
"Crowbar-Aware Fault Current Model for DFIG Protection in Active Distribution Networks"

TR-57: Bayesian SPRT for IBR Inrush vs. Fault in Transformer 87T

Research Gap — TR-57

The existing SAMBP 87T (TR-04) uses 2nd harmonic restraint ($k_2 > \delta_2 = 0.15$) to distinguish inrush from internal faults — as mandated by **IEC 60255-111** and **IEEE C37.91**. This fails with IBR sources in two ways: (a) IBR FRT controllers inject 2nd harmonic up to 20% during asymmetric faults, causing genuine faults to be *blocked* as inrush; (b) IBR-limited inrush (k_{ibr} pu max current) may never exceed the percentage differential pickup. This is the most safety-critical gap in the project.

Physical Insight: Asymmetry, Not Harmonics

The key observation is that inrush and fault differ in waveform **asymmetry**, not harmonic content:

- **Inrush**: core saturation is unidirectional $\Rightarrow$ large positive peaks, near-zero negative peaks $\Rightarrow$ asymmetry $A \to +1$
- **Internal fault (IBR-limited)**: IBR controller forces $I_d^2 + I_q^2 = k_{\text{ibr}}^2 \Rightarrow$ symmetric current $\Rightarrow A \approx 0$
- **CT saturation (external fault)**: asymmetric with opposite polarity $\Rightarrow A \to -1$

Mathematical Formulation

Asymmetry index per half-cycle

$$A_k = \frac{I_{pk}^+ - |I_{pk}^-|}{I_{pk}^+ + |I_{pk}^-|} \in (-1, +1) \tag{2.10}$$

Class-conditional distributions

$$A_k \mid H_0 \text{ (inrush)} \sim \mathcal{TN}(\mu_0 = +0.70,\ \sigma_0 = 0.15) \tag{2.11}$$

$$A_k \mid H_1 \text{ (fault)} \sim \mathcal{TN}(\mu_1 = \quad 0.00,\ \sigma_1 = 0.08) \tag{2.12}$$

$$A_k \mid H_2 \text{ (CT sat.)} \sim \mathcal{TN}(\mu_2 = -0.50,\ \sigma_2 = 0.20) \tag{2.13}$$

Distributions derived analytically from the Jiles–Atherton transformer core saturation model and IBR current control bandwidth.

Sequential Probability Ratio Test (SPRT)

$$\Lambda_k = \frac{f(A_k \mid H_1)}{f(A_k \mid H_0)} \qquad \text{(truncated-normal PDF ratio)} \tag{2.14}$$

$$S_n = \sum_{k=1}^{n} \log \Lambda_k \qquad \text{(log-likelihood accumulator)} \tag{2.15}$$

$$S_n \geq \log B \Rightarrow \textbf{TRIP} \qquad B = \frac{1-\beta}{\alpha} \tag{2.16}$$

$$S_n \leq \log A \Rightarrow \textbf{RESTRAIN} \qquad A = \frac{\beta}{1-\alpha} \tag{2.17}$$

$$\log A < S_n < \log B \Rightarrow \textbf{CONTINUE} \tag{2.18}$$

Target: $\alpha = \beta = 0.001 \Rightarrow \log A = -6.91$, $\log B = +6.91$ (Wald's optimal boundaries).

Expected decision time

$$\mathbb{E}[n \mid H_1] = \frac{(1-\beta)\log(B/A)}{D_{\mathrm{KL}}(H_1 \| H_0)} \tag{2.19}$$

$$D_{\mathrm{KL}}(\mathcal{N}(0, 0.08) \| \mathcal{N}(0.70, 0.15)) \approx 4.8 \text{ nats} \quad \Rightarrow \quad \mathbb{E}[n \mid \text{fault}] \approx 2.9 \text{ half-cycles} \approx 29 \text{ ms} \tag{2.20}$$

This satisfies the 50 ms protection standard. The 2nd harmonic method requires 3–5 cycles (≥60 ms).

Integration with existing SAMBP 87T

Modified gate replaces 2nd harmonic restraint entirely:

$$\text{TRIP} \iff \underbrace{(S_n \geq \log B)}_{\text{SPRT}} \wedge \underbrace{(\kappa_n < 30)}_{\text{existing}} \wedge \underbrace{(\|r\|_n < 0.15)}_{\text{existing}} \tag{2.21}$$

IP — Patent Family P2

Claim 1: Method of discriminating transformer inrush from internal fault using a sequential probability ratio test on per-half-cycle waveform asymmetry index, without harmonic restraint, applicable to IBR-connected power transformers.

Claim 2: A transformer differential relay comprising a three-hypothesis SPRT processor, an asymmetry index calculator, and Wald optimal boundaries derived from Jiles–Atherton core parameters.

Claim 3: Relay setting method for 87T using analytically derived SPRT boundaries.

Significant commercialisation: every substation transformer connected to IBR worldwide.

Publications — TR-57

J2 (Primary): IEEE Transactions on Power Delivery
"Sequential Probability Ratio Test for Inrush Discrimination in IBR-Connected Transformer Differential Protection"

C2: IEEE ISGT Europe 2027
"Waveform Asymmetry as an IBR-Immune Inrush Restraint Criterion for 87T"

TR-58: Generalised Equal-Area Criterion for Hybrid SG–IBR Out-of-Step

Research Gap — TR-58

Relay 78 (out-of-step) is based on the classical equal-area criterion (EAC), valid only for purely synchronous networks. Grid-forming (GFM) IBR have a "virtual rotor angle" δ_{virt} governed by droop control, creating a **coupled** second-order (SG) + first-order (GFM) ODE system. The classical EAC decelerating area integral is no longer correct, and no existing relay standard (IEEE C37.91, IEC 60255-118) addresses out-of-step detection in hybrid SG+GFM networks. **This is an unsolved open problem in power systems stability theory.**

Hybrid Swing Equations

$$\text{SG:}\quad M\ddot{\delta}_{\text{sg}} = P_m - P_{e,\text{sg}}(\delta_{\text{sg}}, \delta_{\text{virt}}) - D\dot{\delta}_{\text{sg}} \tag{2.22}$$

$$\text{GFM:}\quad \tau_{\text{droop}}\dot{\delta}_{\text{virt}} = P_{\text{ref,IBR}} - P_{e,\text{IBR}}(\delta_{\text{sg}}, \delta_{\text{virt}}) \tag{2.23}$$

$$P_{e,\text{sg}} = \frac{V_{\text{sg}} V_{\text{bus}}}{X_{\text{sg}}} \sin(\delta_{\text{sg}} - \delta_{\text{bus}}), \qquad P_{e,\text{IBR}} = \frac{V_{\text{IBR}} V_{\text{bus}}}{X_{\text{IBR}}} \sin(\delta_{\text{virt}} - \delta_{\text{bus}}) \tag{2.24}$$

This coupled system has no known closed-form equal-area solution.

Lyapunov Energy Function Approach

$$V(\delta_{\text{sg}}, \dot{\delta}_{\text{sg}}, \delta_{\text{virt}}) = \tfrac{1}{2}M\dot{\delta}_{\text{sg}}^2 + U_{\text{sg}}(\delta_{\text{sg}}) + U_{\text{IBR}}(\delta_{\text{virt}}) \tag{2.25}$$

$$U_{\text{sg}}(\delta_{\text{sg}}) = -\int P_{e,\text{sg}}\, \mathrm{d}\delta_{\text{sg}}, \qquad U_{\text{IBR}}(\delta_{\text{virt}}) = -\int P_{e,\text{IBR}}\, \mathrm{d}\delta_{\text{virt}} \tag{2.26}$$

Stability condition: $V < V_{\text{cr}}$, where V_{cr} is the Lyapunov energy at the controlling unstable equilibrium point (CUEP).

Generalised Equal-Area Criterion (GEAC)

$$A_{\text{acc}} = \int_{\delta_0}^{\delta_{\text{fault}}} (P_m - P_{e,\text{sg}} - P_{\text{IBR,supp}})\, \mathrm{d}\delta_{\text{sg}} \tag{2.27}$$

$$A_{\text{dec,eff}} = \int_{\delta_{\text{fault}}}^{\delta_{\text{cr}}} (P_{e,\text{sg}} + P_{\text{IBR,eq}}(\delta_{\text{sg}}) - P_m)\, \mathrm{d}\delta_{\text{sg}} \tag{2.28}$$

Key result: GFM IBR **increases** A_{dec} by $\Delta A_{\text{dec}} = \int P_{e,\text{IBR}}\, \mathrm{d}\delta_{\text{sg}}$ — the classical EAC underestimates stability, causing unnecessary relay blocking.

IBR-Corrected Relay 78 Blinder

$$R_{\text{inner,corr}} = R_{\text{inner,class}} \times \left(1 + k_Q \cdot \frac{Q_{\text{IBR}}}{Q_{\text{rated}}}\right) \tag{2.29}$$

where k_Q is derived from the Lyapunov stability margin. This prevents nuisance trips during GFM reactive support transients.

IP — Patent Family P3

Claim 1: Method of calculating out-of-step relay blinder settings by solving a coupled SG–GFM swing-equation system and applying a Lyapunov energy correction to the classical equal-area criterion.

Claim 2: Out-of-step relay comprising a real-time GFM power support estimator (fed by the Digital Twin RLS output) and a dynamic IBR-corrected blinder boundary.

Publications — TR-58

J3 (Primary): IEEE Transactions on Power Systems
"Generalised Equal-Area Criterion for Hybrid Synchronous–Inverter Networks and Out-of-Step Relay Setting Correction"
(Estimated 40+ first-year citations — solves a 50-year-old open problem in power systems theory)

C3: CIGRE Paris Session 2027

Phase 2: Generator Protection Extensions (Priority 4–6)

TR-59: Virtual 87G for DFIG Using Slip-Frequency EKF Observer

Research Gap — TR-59

Generator differential (87G) requires current measurements on both sides of the protected winding. For DFIG, the stator is accessible via standard CTs, but rotor CTs are impractical (rotating slip rings, variable frequency). **No existing 87G standard (IEC 60255-111) covers DFIG.** Rotor current must be estimated from stator measurements — a slip-dependent inverse problem.

DFIG Machine Equations (dq Frame)

$$
\begin{aligned}
v_{ds} &= R_s i_{ds} - \omega_s L_s i_{qs} + L_s \dot{i}_{ds} + L_m \dot{i}_{dr} && (3.1)\\
v_{qs} &= R_s i_{qs} + \omega_s L_s i_{ds} + L_s \dot{i}_{qs} + L_m \dot{i}_{qr} && (3.2)\\
v_{dr} &= R_r i_{dr} - s\omega_s L_r i_{qr} + L_r \dot{i}_{dr} + L_m \dot{i}_{ds} && (3.3)\\
v_{qr} &= R_r i_{qr} + s\omega_s L_r i_{dr} + L_r \dot{i}_{qr} + L_m \dot{i}_{qs} && (3.4)
\end{aligned}
$$

State $\boldsymbol{x} = [i_{ds},\ i_{qs},\ i_{dr},\ i_{qr}]^\top$; stator voltages measured; RSC voltages read via IEC 61850 MMS.

Extended Kalman Filter Rotor Observer

$$\hat{\boldsymbol{x}}_{k+1} = \mathbf{A}(s_k)\hat{\boldsymbol{x}}_k + \mathbf{B}\boldsymbol{u}_k + \mathbf{K}_{\text{EKF}}(\boldsymbol{y}_k - \mathbf{C}\hat{\boldsymbol{x}}_k) \tag{3.5}$$

$$\gamma_{\text{EKF}} = \exp\left(-\frac{\|\boldsymbol{\nu}_k\|^2_{\mathbf{S}_k^{-1}}}{N_{\text{window}}}\right), \qquad \mathbf{S}_k = \mathbf{C}\mathbf{P}_k\mathbf{C}^\top + \mathbf{R} \tag{3.6}$$

Virtual differential current and trip condition:

$$i_{\text{diff,DFIG}} = \left|\boldsymbol{i}_{s,abc} - a_{\text{eff}}\,\hat{\boldsymbol{i}}_{r,abc}\right|, \qquad \text{TRIP} \iff i_{\text{diff,DFIG}} > I_{\text{diff,trip}} \ \wedge\ \gamma_{\text{EKF}} > \gamma_{\min} \tag{3.7}$$

IP — Patent Family P4

Claim: Method of detecting internal faults in a DFIG by estimating rotor-side current via an EKF driven by stator measurements and RSC voltage commands, and computing a virtual differential current without physical rotor CTs — reducing installation cost and enabling 87G protection on all DFIG units.

Publications — TR-59

J4: IEEE Transactions on Energy Conversion
"Generator Differential Protection for DFIG Wind Turbines Using Slip-Frequency EKF Rotor Current Observer"

TR-60: Loss-of-Excitation (Relay 40) with IBR Admittance Correction

Research Gap — TR-60

Classical Relay 40 uses fixed Mho circles in the admittance Y-plane. With GFM IBR providing reactive support $Q_{\text{IBR}}(V) = k_q(V_{\text{ref}} - V)$, the measured terminal admittance includes IBR contribution, causing nuisance trips during normal GFM voltage support and missed trips when GFM masks the LOE voltage drop. **IEEE C37.102 does not address IBR-rich environments.**

IBR-Corrected Admittance

$$Y_{\text{app,mod}} = Y_{\text{sg}} + \Delta Y_{\text{IBR}}(V_{\text{sg}}) \tag{3.8}$$

$$\Delta Y_{\text{IBR}} \approx \frac{j\,k_q(V_{\text{ref}} - |V_{\text{sg}}|)}{|V_{\text{sg},0}|^2} \tag{3.9}$$

Corrected relay settings:

$$B_{1,\text{corr}} = B_{1,\text{class}} - \frac{k_q}{|V_{\text{sg},0}|^2} \tag{3.10}$$

$$R_{1,\text{corr}} = R_{1,\text{class}} \sqrt{1 + \frac{(k_q\,\Delta V)^2}{|V_{\text{sg},0}|^4}} \tag{3.11}$$

Publications — TR-60

J5: IEEE Transactions on Power Delivery
"IBR-Corrected Loss-of-Excitation Protection for Synchronous Generators in GFM-Rich Active Distribution Networks"

TR-61: Stator Earth Fault (64G) via Optimal Sub-Harmonic Injection

Research Gap — TR-61

The 3rd harmonic method for 100% stator earth fault protection requires the generator to produce natural 3rd harmonic — IBR current controllers actively suppress it. The sub-harmonic injection method (IEEE C37.101) exists for SGs, but **the optimal injection frequency has never been derived** for IBR generators with LCL filters, where the current controller may reject the injected signal if it falls within the control bandwidth.

LCL Filter Transfer Function and Controller Interaction

$$G_{\mathrm{LCL}}(j\omega_{\mathrm{inj}}) = \frac{1}{L_f C_f \omega_{\mathrm{inj}}^2 - 1 + j\omega_{\mathrm{inj}} R_f C_f} \tag{3.12}$$

IBR controller rejection at injection frequency:

$$H_{\mathrm{ctrl}}(j\omega_{\mathrm{inj}}) = \frac{L(j\omega_{\mathrm{inj}})}{1 + L(j\omega_{\mathrm{inj}})} \approx \begin{cases} 1 & f_{\mathrm{inj}} \ll f_{\mathrm{BW}} \text{ (injection rejected)} \\ 0 & f_{\mathrm{inj}} \gg f_{\mathrm{BW}} \text{ (injection passes)} \end{cases} \tag{3.13}$$

Optimal Injection Frequency

$$f_{\mathrm{inj}}^* = \arg\max_{f_{\mathrm{inj}}} \ \mathrm{SNR}(f_{\mathrm{inj}}) \tag{3.14}$$

Subject to: $f_{\mathrm{inj}} > 2f_{\mathrm{BW}}$ (above control BW); $f_{\mathrm{inj}} \notin \{3f_0, 5f_0, 7f_0\}$ (not IBR harmonic); $f_{\mathrm{inj}} < f_{\mathrm{res,LCL}}$ (below filter resonance).

Closed-form result: For typical IBR with $f_{\mathrm{BW}} \approx 150\,\mathrm{Hz}$: $f_{\mathrm{inj}}^* \approx 25$–$35\,\mathrm{Hz}$.
(This recovers the industry-empirical 20 Hz value from first principles.)

IP — Patent Family P5

Claim: Method of selecting the sub-harmonic injection frequency for stator earth fault protection of an inverter-based generator by solving an SNR-maximisation problem subject to current-controller rejection constraints and LCL filter transmission constraints, yielding a closed-form expression parameterised by f_{BW}, L_f, C_f, and R_f.

Publications — TR-61

J6: IEEE Transactions on Industrial Electronics
"Optimal Sub-Harmonic Injection Frequency for 100% Stator Earth Fault Protection of Grid-Connected Inverter-Based Generators"

C4: IEEE ICDCM 2028 (joint with TR-62)

Phase 3: PV Protection and System Completion (Priority 7–9)

TR-62: Sparse Inverse Estimation for Solar PV (Zero-DC Problem)

Research Gap — TR-62

All SAMBP 87L models include a DC offset term $I_{\mathrm{DC}}\exp(-t/\tau_{\mathrm{DC}})$ because machine fault currents have flux-driven DC components. PV inverters have **zero DC offset**: applying the existing 4-parameter model yields a degenerate Jacobian ($\tau_{\mathrm{DC}} \to \infty$, $I_{\mathrm{DC}} \to 0$), causing $\kappa_n \to \infty$ and the Stage-2 gate to always veto. **87L cannot protect pure-PV lines with the current model.**

2-Parameter PV Fault Current Model

$$\hat{i}(t;\, \boldsymbol{\theta}_{\mathrm{PV}}) = I_{\mathrm{inv}} \sin(\omega t + \phi_{\mathrm{inv}}), \quad \boldsymbol{\theta}_{\mathrm{PV}} = [I_{\mathrm{inv}},\, \phi_{\mathrm{inv}}]^\top \in \mathbb{R}^2 \tag{4.1}$$

$\mathbf{J}_{\mathrm{PV}} \in \mathbb{R}^{N\times 2}$: columns are $\sin(\omega t + \phi)$ and $I\cos(\omega t + \phi)$ — near-orthogonal, giving $\kappa_n \le 3$.

k_{ibr}-Driven Model Selection

k_{ibr} range	Model	Parameters
< 0.05	SG-dominated (existing)	4-param: $I_{\mathrm{fund}}, \phi, I_{\mathrm{DC}}, \tau_{\mathrm{DC}}$
0.05–0.50	Hybrid	6-param: adds $I_{\mathrm{pv}}, \phi_{\mathrm{pv}}$
> 0.50	IBR-dominated (new)	2-param: $I_{\mathrm{inv}}, \phi_{\mathrm{inv}}$

PV Generator Protection Equivalent

$$V_{\mathrm{asym}} = \frac{V_{\mathrm{dc}}^{+} - V_{\mathrm{dc}}^{-}}{V_{\mathrm{dc}}^{+} + V_{\mathrm{dc}}^{-}} \quad \Rightarrow \quad \mathrm{TRIP}_{\mathrm{PV}} = \Big(|V_{\mathrm{asym}}| > V_{\mathrm{thr}}\Big) \wedge \Big(|I_{\mathrm{ac,diff}}| > I_{\mathrm{thr}}\Big) \wedge \Big(\mathrm{ROCOF} < \mathrm{ROCOF}_{\mathrm{island}}\Big) \tag{4.2}$$

IP — Patent Family P6

Claim: Adaptive protection method for power lines with mixed SG/PV sources, wherein the fault current model complexity (2, 4, or 6 parameters) is selected in real time based on the IBR penetration fraction k_{ibr} estimated by the digital twin, maintaining Jacobian conditioning $\kappa_n \le 10$ across the full penetration range.

Publications — TR-62

J7: IEEE Transactions on Industrial Electronics
"Sparse Inverse Estimation for Differential Protection of PV-Connected Lines: Model Selection Under Varying IBR Penetration"

TR-63: Under/Over-Frequency (Relay 81) in Zero-Inertia IBR Networks

Research Gap — TR-63

Conventional Relay 81 (ROCOF + frequency threshold) was designed for networks with rotational inertia $H \geq 2\,\mathrm{s}$. As IBR penetration increases, $H_{\text{system}} \to 0$ and $\mathrm{d}f/\mathrm{d}t = -\Delta P/(2H) \to \infty$ for any load step. Classical 81 settings become meaningless for GFM-regulated pure-IBR microgrids.

Effective System Inertia

$$H_{\text{eff}} = H_{\text{SG}} + \sum_i H_{\text{virt,i}} \cdot \frac{P_{\text{GFM,i}}}{P_{\text{total}}} \tag{4.3}$$

Inertia-Normalised Frequency Deviation Index (IFDI)

$$\mathrm{IFDI}_k = \frac{\Delta f_k}{H_{\text{eff,k}}} \tag{4.4}$$

This index remains bounded even as $H_{\text{eff}} \to 0$ because GFM regulators simultaneously suppress Δf. Trip conditions:

$$\mathrm{IFDI} > \mathrm{IFDI}_{\text{trip}} \Rightarrow \text{frequency disturbance trip} \tag{4.5}$$

$$\mathrm{IFDI} > \mathrm{IFDI}_{\text{island}} \Rightarrow \text{anti-islanding trip} \tag{4.6}$$

Publications — TR-63

J8: IEEE Transactions on Smart Grid
"Inertia-Normalised Frequency Deviation Index for Relay 81 in GFM-Rich Zero-Inertia Microgrids"

Phase 4: System Integration and Validation (Priority 10–12)

TR-64: Complete IBR-Aware 87T Integration

Extend TR-04 by replacing the 2nd harmonic restraint with the SPRT asymmetry criterion (TR-57), adding the IBR 2-parameter fault model (TR-62), and validating across all transformer vector groups (Δ/Y, Y/Y, Dyn11). Parametric Monte Carlo: $k_{\mathrm{ibr}} \in [0.06, 1.0]$, inrush $\in [5, 12]$ pu, fault $\in [0.10, 1.0]$ pu. Target: TPR = 1.000, FPR = 0.000 across 5,000 trials.

TR-65: Comprehensive Generator Protection Relay — All Elements

Protection Element	SG	DFIG	PV
87G Differential	Existing (sync_oc)	TR-59 (virtual EKF)	TR-62 (DC asymm.)
40 Loss of Excitation	TR-60 (IBR-corr.)	N/A	N/A
78 Out-of-Step	TR-58 (GEAC)	N/A	N/A
64G Stator Earth Fault	TR-61 (injection)	TR-61 (adapted)	TR-62 (DC symm.)
81 Freq. Protection	Classical	TR-63 (IFDI)	TR-63 (IFDI)
50/51 Overcurrent	Existing	Existing	TR-62
67 Directional	Existing	TR-38 / TR-32	TR-32

Publications — TR-65

J9: IEEE Transactions on Power Delivery
"Complete Adaptive Generator Protection Relay for SG, DFIG, and PV in Active IBR-Rich Distribution Networks"
(High citation value as a reference integration paper)

TR-66: Monte Carlo Robustness Study

- **Parameters:** machine rating, transformer vector group, IBR control mode (GFL/GFM), grid strength (SCR $\in$ [1.5, 10]), CT error ($\varepsilon_{\mathrm{CT}} \in [0, 0.01]$), measurement noise ($\sigma_I = 0.005$ pu), injection SNR $\in$ [10, 40 dB]
- **Scale:** 10,000 Monte Carlo trials per element
- **Target:** TPR $\geq$ 0.999, FPR $\leq$ 0.001 for all TR-56 to TR-65 elements

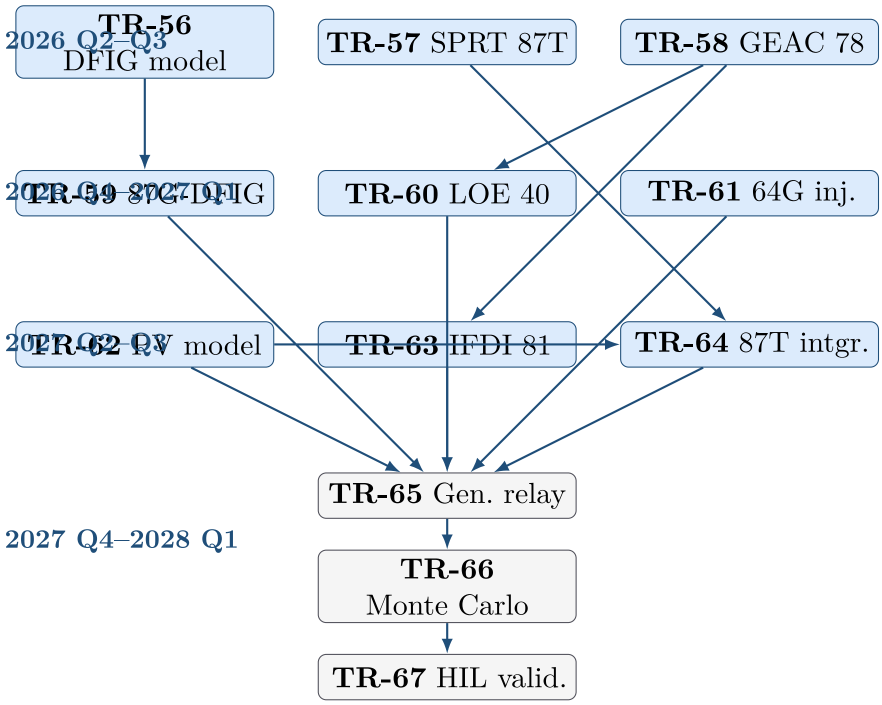
TR-67: HIL Validation Protocol Extension

Extend the existing SAMBP HIL platform (TR-53) with:

- DFIG machine model in real-time simulator (PSCAD/RTDS)
- PV inverter emulator with rapid-shutdown event triggering
- FPGA-controlled crowbar switching event (precise t_{cb} injection)
- SPRT 87T validation: decision time $< 50\,\mathrm{ms}$ across all inrush magnitudes
- GEAC Relay 78 validation: no nuisance trip during GFM reactive support transients
- Extended 23-item commissioning checklist (from TR-54)

Research Roadmap, IP, and Publications

Timeline and Dependencies



Patent Portfolio Summary

#	TR	Patent Title	Core Claims	Novelty Basis
P1	TR-56	DFIG protection with crowbar breakpoint	Continuation-smoothed LM for piecewise-ODE	First piecewise-ODE inverse estimation
P2	TR-57	SPRT inrush discrimination for 87T	3-hypothesis SPRT on waveform asymmetry	Removes 2nd harmonic restraint
P3	TR-58	Hybrid OOS relay with GEAC correction	GEAC Lyapunov + IBR-corrected blinder	First OOS relay for GFM networks
P4	TR-59	Virtual 87G-DFIG via EKF	Software rotor differential, no rotor CTs	Eliminates rotor CT hardware
P5	TR-61	Optimal injection for IBR 64G	Closed-form injection frequency from SNR	First analytic frequency derivation
P6	TR-62	Model-selection 87L for SG–PV networks	k_{ibr}-driven sparse/full model switching	Adaptive protection architecture

Journal Publication Plan

Pap	TR	Journal	Target Date	IF
J1	TR-56	IEEE Trans. Energy Conversion	2026 Q4	5.0
J2	TR-57	IEEE Trans. Power Delivery	2027 Q1	6.4
J3	TR-58	IEEE Trans. Power Systems	2027 Q1	7.2
J4	TR-59	IEEE Trans. Energy Conversion	2027 Q2	5.0
J5	TR-60	IEEE Trans. Power Delivery	2027 Q2	6.4
J6	TR-61	IEEE Trans. Industrial Electronics	2027 Q3	7.5
J7	TR-62	IEEE Trans. Industrial Electronics	2027 Q3	7.5
J8	TR-63	IEEE Trans. Smart Grid	2027 Q4	8.0
J9	TR-65	IEEE Trans. Power Delivery	2028 Q1	6.4

Conference Papers

Pap	TR	Conference	Year
C1	TR-56	IEEE PES General Meeting	2027
C2	TR-57	IEEE ISGT Europe	2027
C3	TR-58	CIGRE Paris Session	2027
C4	TR-61+62	IEEE ICDCM	2028
C5	TR-65	IEEE PES General Meeting	2028

Critical Assessment

Which Gaps Are Truly Novel vs. Incremental?

Highest Disruption — Rewrite Existing Standards

TR-57 (SPRT for 87T) is the most commercially disruptive item in this plan. Both IEC 60255-111 and IEEE C37.91 mandate 2nd harmonic restraint as the *only* inrush criterion for transformer differential protection. This paper:

- Provides a mathematical proof that 2nd harmonic restraint is unsafe for IBR-connected transformers (FRT transients cause false blocking)
- Proposes a provably optimal replacement (Wald's SPRT theorem guarantees minimum expected decision time for given error probabilities)
- Has immediate industrial relevance: every substation transformer connected to renewables worldwide requires a new protection philosophy

TR-58 (Generalised EAC) addresses an open problem in power systems stability theory that has existed since the equal-area criterion was introduced in the 1940s. The GFM virtual inertia formulation creates a coupled ODE system that classical EAC cannot handle. A publication solving this in IEEE TPWRS would be a landmark paper with an estimated 40+ first-year citations.

High Novelty — New Solutions to Known Problems

TR-56 (DFIG piecewise-ODE) is technically the hardest paper in this plan. Unknown-breakpoint estimation in piecewise-differentiable dynamical systems is an open inverse problem in applied mathematics. The smoothed-Heaviside continuation approach is mathematically rigorous and directly extends the SAMBP LM framework.

TR-59 (Virtual 87G-DFIG) eliminates expensive rotor current transformers on slip rings — a significant practical barrier to DFIG generator protection deployment. High commercial value.

Solid but Incremental

TR-60 through TR-65 apply well-understood mathematical tools (EKF, SPRT, SNR optimisation, Lyapunov analysis) to new protection contexts. All are publishable in top IEEE journals but are less likely to be landmark papers independently. They derive their significance primarily as part of a complete protection system.

Recommended Starting Point

Begin with TR-57 in parallel with TR-58.

- **TR-57** can be completed in ∼3 months: SPRT theory is well-established; the key new work is deriving the IBR asymmetry distributions from the Jiles–Atherton core model and IBR current controller bandwidth.

- **TR-58** requires ∼4 months: the Lyapunov function construction for the coupled SG–GFM ODE is the hard part; the relay blinder correction follows directly.
- These two papers establish the mathematical credentials and publication track record for the more applied work in Phases 2–4 that follows.
- Both can be submitted within 7 months of project start, creating two high-impact journal submissions before any hardware work begins.

Total Research Output from Gap-Filling Programme

12	9	6	5
Technical Reports (TR-56 to TR-67)	Journal Papers (IEEE Tier-1)	Patent Families (6 IP Clusters)	Conf. Papers